Practice workbook · Volume formulas for cones, pyramids and prisms using trig

Writing the volume of a cone, pyramid or prism in terms of one variable by using trig to link its dimensions, then solving or rearranging.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-19

1. Question 9 · part a · 5 marks

2025 NOV · Paper 01

Worked-example retry · Attempt part a only.

9

Diagram **NOT** accurately drawn

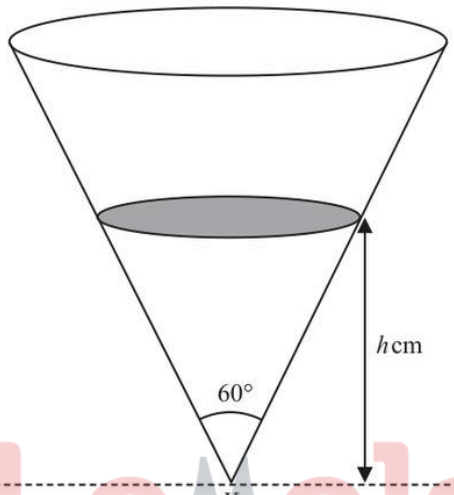


Figure 2

Figure 2 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The vertical angle of the cone is 60°
Initially the cone is empty.

At time $t = 0$ liquid starts to fill the cone at a constant rate of $0.2\,\text{cm}^3/\text{s}$
At time t seconds after the liquid starts to fill the cone, the height of the liquid is h cm above X

(a) Show that $h = \sqrt[3]{\frac{9t}{5\pi}}$

(5)

The surface area of the liquid, shown shaded in Figure 2, is increasing at a constant rate of $p\,\text{cm}^2/\text{s}$ when $t = 6$

(b) Find, to 3 significant figures, the value of p

(8)

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1. Question 9 · part a · 5 marks · Working and answer

Working and answer

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2. Question 10 · part e · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

10

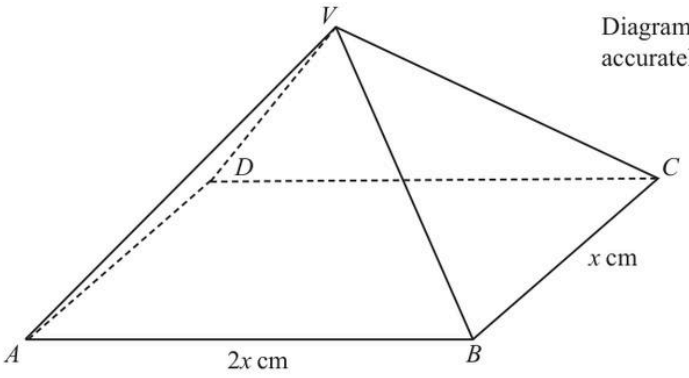


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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2. Question 10 · part e · 2 marks

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Further practice · Attempt part e only.

Question 10 continued

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2. Question 10 · part e · 2 marks

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Further practice · Attempt part e only.

Question 10 continued

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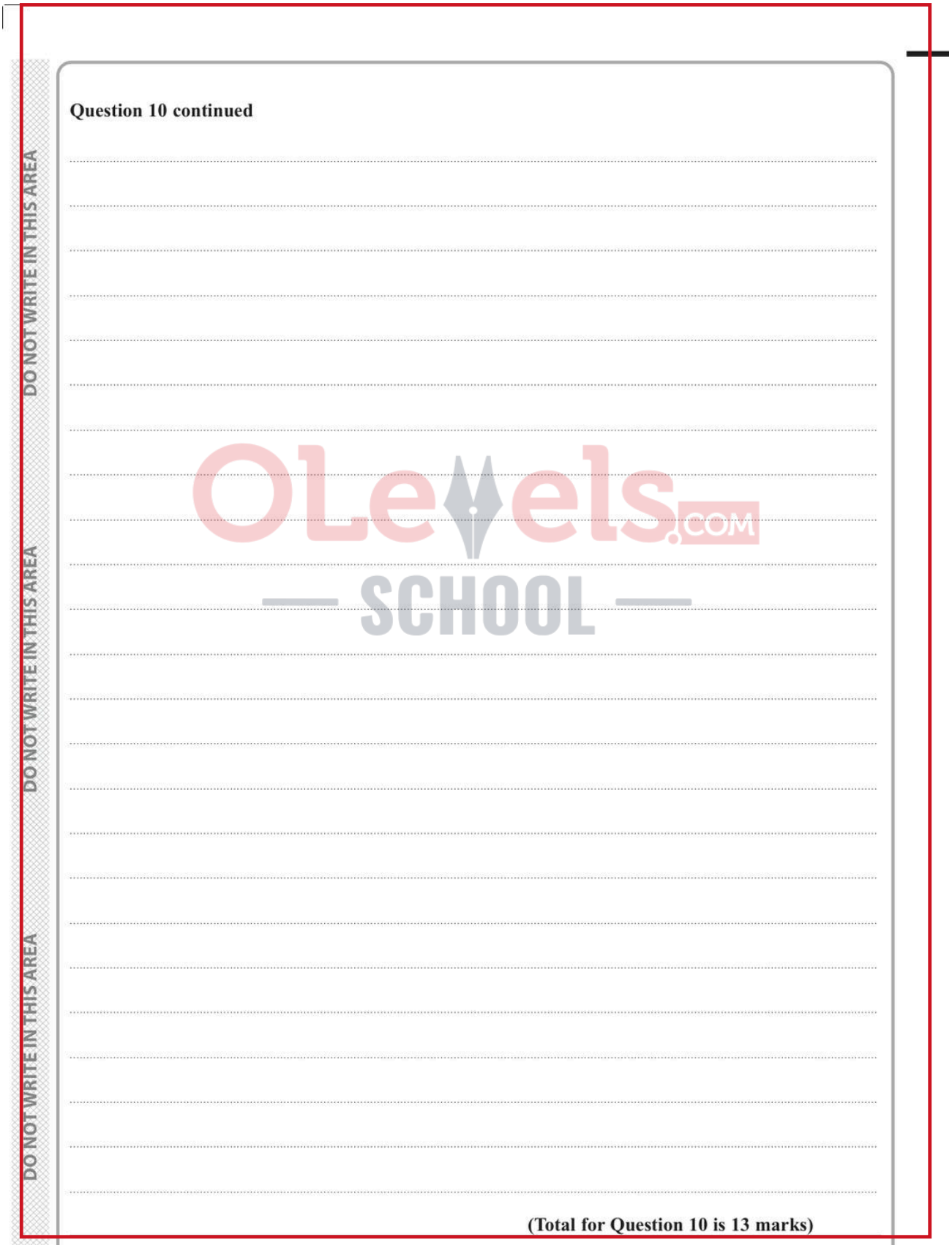
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2. Question 10 · part e · 2 marks

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Further practice · Attempt part e only.



Question 10 continued

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(Total for Question 10 is 13 marks)

2. Question 10 · part e · 2 marks · Working and answer

Working and answer

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3. Question 6 · part d · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

6

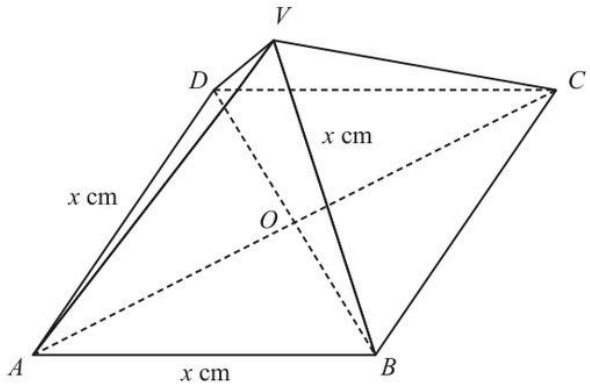


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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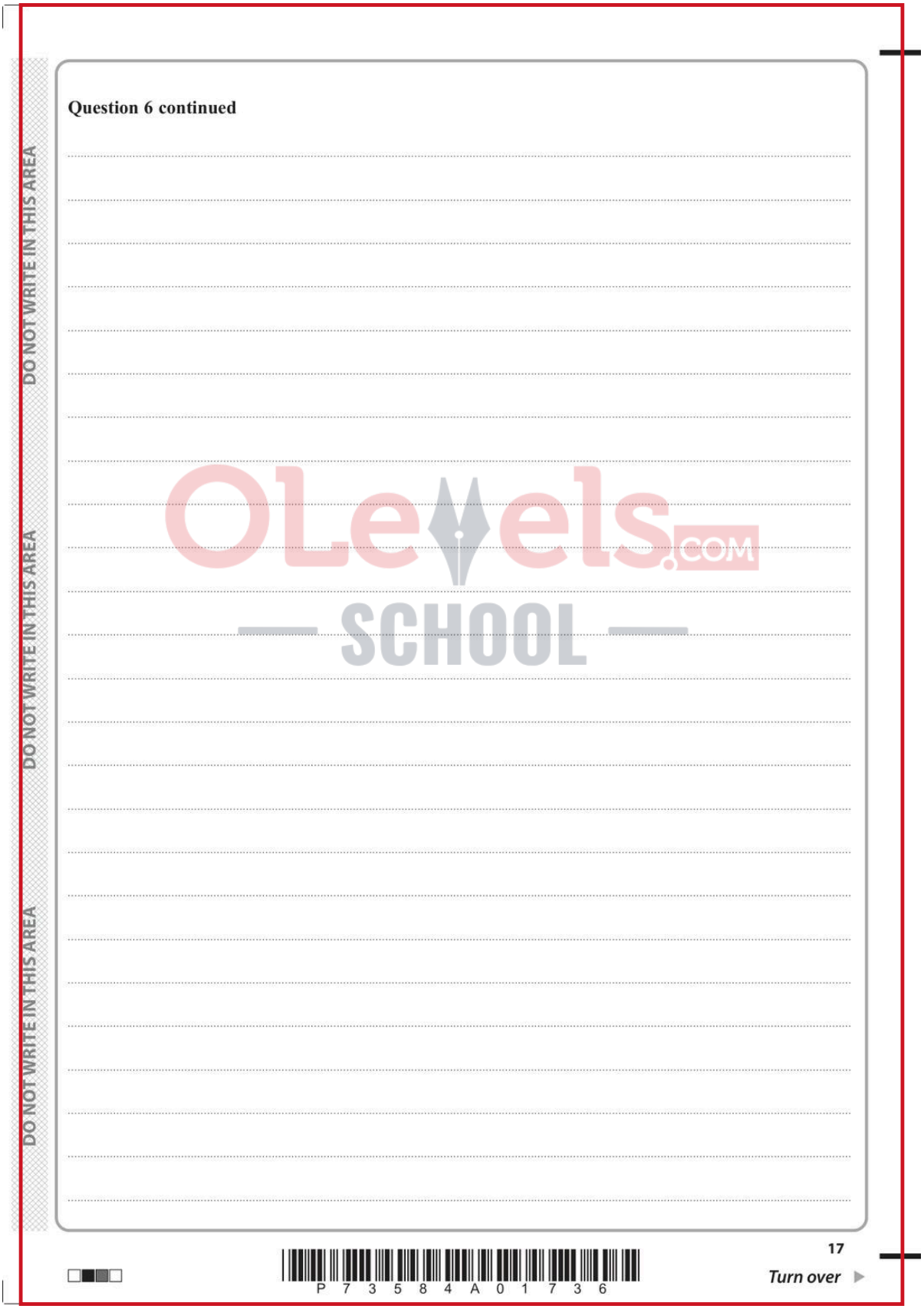
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3. Question 6 · part d · 3 marks

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Further practice · Attempt part d only.



Question 6 continued

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Turn over ▶

3. Question 6 · part d · 3 marks

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Further practice · Attempt part d only.

Question 6 continued

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3. Question 6 · part d · 3 marks

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Question 6 continued

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(Total for Question 6 is 11 marks)

3. Question 6 · part d · 3 marks · Working and answer

Working and answer

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4. Question 4 · part a · 3 marks

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Further practice · Attempt part a only.

4

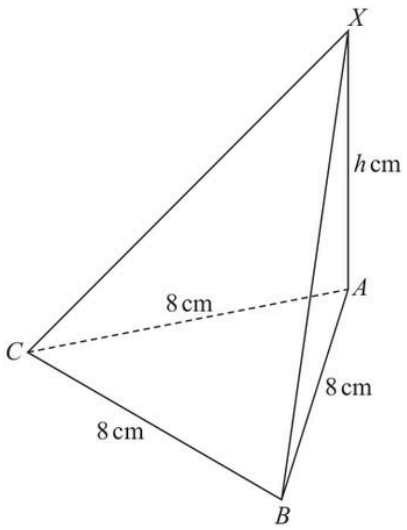


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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4. Question 4 · part a · 3 marks · Working and answer

Working and answer

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5. Question 6 · part a · 3 marks

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Further practice · Attempt part a only.

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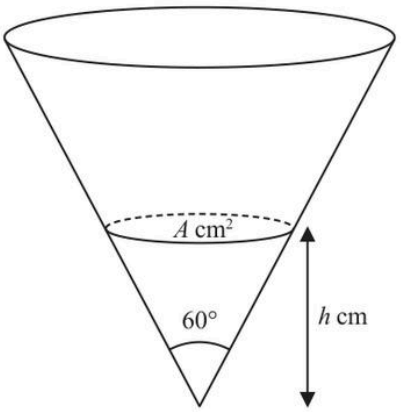


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a container in the shape of a right circular cone.
The container is fixed with its axis of symmetry vertical.
The vertical angle of the container is 60° as shown in the diagram.

At time t seconds, $t > 0$, the height of oil in the container is h cm and the volume of oil in the container is V cm^3

(a) Show that $V = \frac{1}{9}\pi h^3$

(3)

At time t seconds the surface area of oil in the container is A cm^2, as shown in Figure 4

Oil is dripping out of the bottom of the container at a constant rate of 4 cm^3/s.

(b) Find the exact rate of change, in cm^2/s, of the surface area of oil in the container when $h = 24$

(8)

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5. Question 6 · part a · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a · 2 marks

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Further practice · Attempt part a only.

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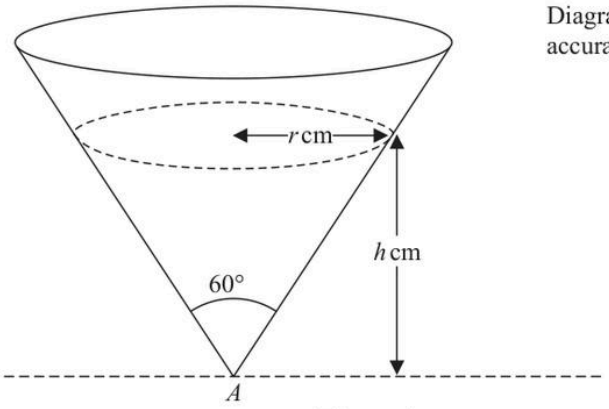


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The cone is inverted and contains liquid, which is dripping out of a small hole at the vertex A of the cone at a constant rate of $0.9\,\text{cm}^3/\text{s}$.

At time t seconds after the liquid starts to drip from the cone, the height of the liquid is h cm above A. The volume of liquid in the cone at time t seconds is $V\,\text{cm}^3$

The vertical angle of the cone is $60°$

(a) Show that $V = \frac{1}{9}\pi h^3$ **(2)**

(b) Find, in cm/s to 3 significant figures, the rate at which the height of the liquid is decreasing when the height of the liquid in the cone above the vertex is 1.2 cm. **(4)**

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6. Question 6 · part a · 2 marks · Working and answer

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7. Question 6 · part a · 1 mark

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Further practice · Attempt part a only.

6

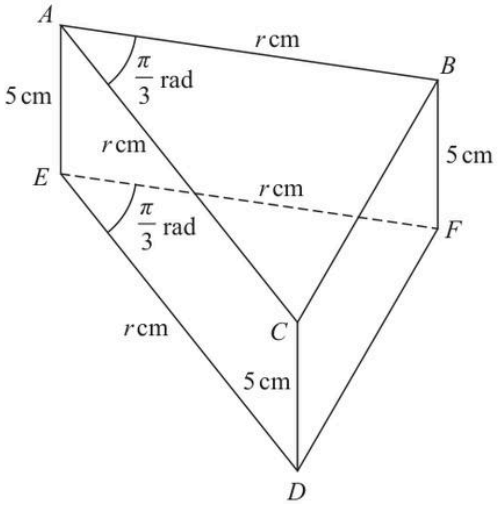


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right triangular prism $ABCDEF$. A cross section ABC of the prism is a triangle in which $AB = AC = r\,\text{cm}$ and $\angle CAB = \frac{\pi}{3}$ radians.

In the prism

$$AE = BF = CD = 5\,\text{cm} \qquad ED = EF = r\,\text{cm} \text{ and } \angle DEF = \frac{\pi}{3} \text{ radians}$$

(a) Show that the volume of the prism is $\frac{5\sqrt{3}}{4} r^2\ \text{cm}^3$

(1)

The volume of the prism is increasing in such a way that the size of $\angle CAB$ and the size of $\angle DEF$ remain constant and the length of AE, the length of BF and the length of CD remain constant.
The lengths of AB, AC, ED and EF are each increasing at a constant rate of $0.2\,\text{cm/s}$

(b) Find the exact rate of increase, in cm^3/s, of the volume of the prism when the area of the rectangular face $BCDF$ is $60\,\text{cm}^2$

(5)

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7. Question 6 · part a · 1 mark · Working and answer

Working and answer

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